Deriving the Chebyshev Distance Using Limits

For a vector $z = (z_1, \ldots, z_n) \in \mathbb{R}^n$, with $n \geq 1$, the ℓ_p norm is

$$\|z\|_p = \left(\sum_{i=1}^{n} |z_i|^p \right)^{1/p}, \qquad 1 \leq p < \infty.$$

We will show that its limit as $p \to \infty$ is the largest component magnitude.

Simple proof. Let $M = \max_{1 \leq i \leq n} |z_i|$. At least one term in the sum equals M^p, and every term is at most M^p. Hence

$$M^p \leq \sum_{i=1}^{n} |z_i|^p \leq n M^p.$$

Taking the pth root gives

$$M \leq \|z\|_p \leq n^{1/p} M.$$

Since

$$\lim_{p \to \infty} n^{1/p} = \lim_{p \to \infty} e^{(\log n)/p} = 1,$$

the squeeze theorem yields

$$\boxed{\lim_{p \to \infty} \|z\|_p = \max_{1 \leq i \leq n} |z_i| = \|z\|_\infty.}$$

This argument also covers $z = 0$, when both bounds are zero.

Chebyshev distance. Applying the result to $z = x - y$ gives the ℓ_∞ (Chebyshev) distance:

$$\boxed{d_\infty(x, y) = \lim_{p \to \infty} \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p} = \max_{1 \leq i \leq n} |x_i - y_i|.}$$

Note. The ℓ_∞ norm selects a component with maximum magnitude and returns that magnitude, not its sign or index. For distance, it selects the largest absolute coordinate difference. Ties do not affect the value.